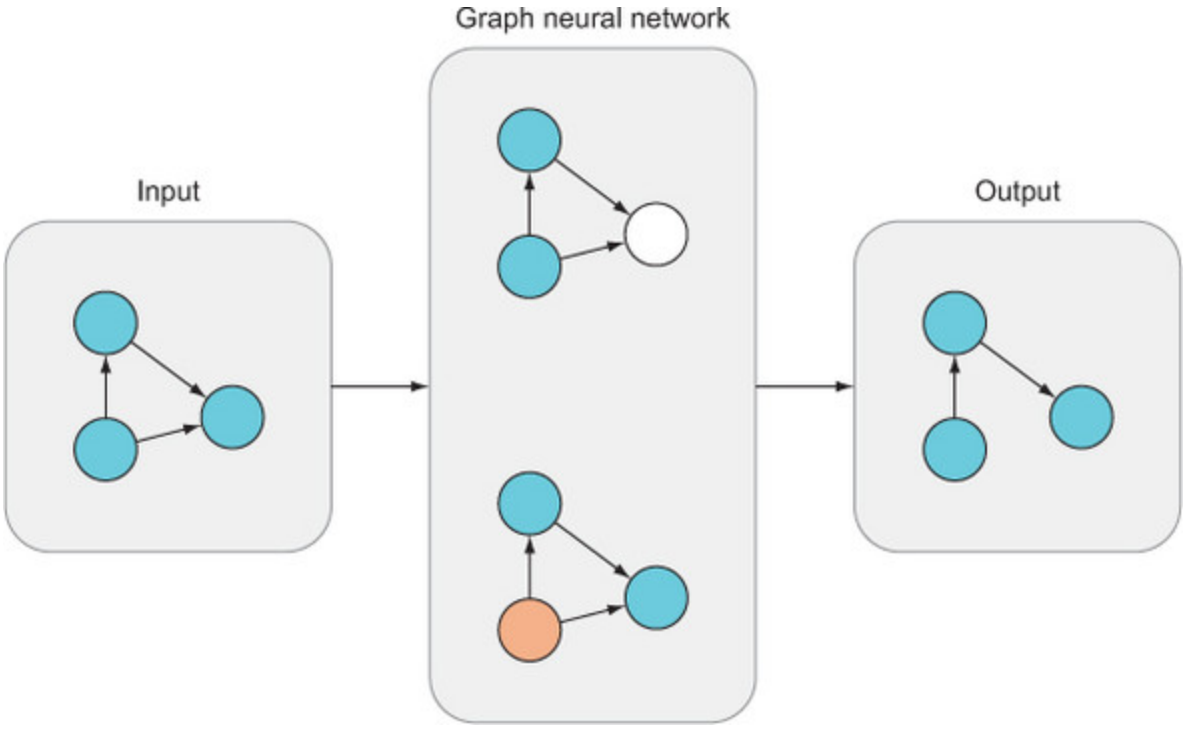


Figure 10.2. A graph neural network can directly operate on a graph, compute over the nodes and edges, and return an updated graph. In this example, the graph neural network decides to remove the edge connecting the bottom two nodes. This is an abstract example, but the nodes could represent real world variables and the arrows represent causal direction, so the algorithm would be learning to infer causal pathways between variables.



Self-attention models (SAMs) can be used to construct graph neural networks, but our goal is not to operate on explicitly graph-structured data; we will instead be working with image data, as usual, but we will use a self-attention model to learn a graph representation of the features within the image. In a sense, we hope the SAM will convert a raw image into a graph structure, and that the graph structure it constructs will be somewhat interpretable. If we train a SAM on a bunch of images of people playing basketball, for example, we might hope it learns to associate the people with the ball, and the ball with the basket. That is, we want it to learn that the ball is a node, the basket is a node, and the players are nodes and to